

ASHITA KUMARI

Computer Science Student • Software Development | AI & Machine Learning

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[GitHub](#) • [LinkedIn](#) • [Portfolio](#)

EDUCATION

Institute of Business Management (IoBM), Karachi

Bachelor of Science in Computer Science • 2024 – Present

CGPA: 3.59 / 4.00

PROJECTS

SkillSwap – Peer Skill Exchange Platform | Python, Flask, MySQL, JavaScript

- Built a full-stack platform where users create profiles and trade skills with each other, with authentication, session-based login, and full CRUD through a Flask REST backend.
- Designed a normalized MySQL schema with foreign keys and indexed queries, connected to a JavaScript frontend for dynamic page updates.

Context-Aware RAG Chatbot | Python, LangChain, Vector Embeddings

- Built a Retrieval-Augmented Generation chatbot that answers questions directly from uploaded documents instead of relying on the model's own memory.
- Set up document chunking and vector embeddings for semantic search, plus a multi-turn memory layer so conversations stay coherent across a session.

AI-Powered Resume & Career Assistant | Java, Google Gemini API

- Developed a Java desktop app that generates resumes and cover letters through Google's Gemini API, replacing manual drafting with a single request-response flow.
- Built the request-response pipeline around encapsulated, reusable classes so new document types can be added without touching existing logic.

Customer Churn Prediction – ML Pipeline | Python, scikit-learn, Pandas

- Built an end-to-end ML pipeline with scikit-learn's Pipeline API, covering imputation, encoding, scaling, and training in one reproducible flow.
- Compared Logistic Regression and Random Forest with hyperparameter tuning, adjusting class weights to improve F1-score on an imbalanced dataset.

University Timetable & CPU Scheduling System | Python, Flask, REST API

- Built a web-based scheduling system with Flask Blueprints in an MVC structure, exposing REST endpoints for full CRUD on courses and time slots.
- Added conflict-detection logic that flags overlapping bookings automatically and generates a clash-free timetable.

Auto-Tagging Support Tickets with LLMs | Python, NLP, Prompt Engineering

- Built an NLP pipeline that classifies and tags incoming support tickets using LLM prompts, covering 3+ ticket categories without manual sorting.
- Structured the prompts around few-shot examples per category to keep classification consistent across similarly worded tickets.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C

Web Development: Flask, REST APIs, HTML5, CSS3, Jinja2

AI & Machine Learning: LangChain, scikit-learn, RAG, NLP, Prompt Engineering, LLMs, Vector Embeddings

Databases: MySQL, SQLite

Developer Tools: Git, GitHub, Postman, VS Code, Jupyter Notebook

LEADERSHIP & ACTIVITIES

- Student Ambassador — School of Leadership
- Promotion Lead — Model United Nations
- Member — Tech Visionary Society, IoBM
- Member — SWAT Society, IoBM
- Participant — Young Leaders Conference (6-day leadership program)

CERTIFICATIONS & ACHIEVEMENTS

IBM SkillsBuild — Artificial Intelligence Fundamentals: machine learning applications, AI ethics, and foundational AI concepts.

Ascend Launchpad 2.0, CSR Forum Pakistan: designed data-driven solutions for urban community challenges in a competitive 4-week program.